Namburi Hemanth Akhiland Kumar

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SKILLS

- **Languages**: C++, Python, SQL, JavaScript
- **AI/ML**: PyTorch, Torchvision, RAG, NLP, CNNs, ResNet18, transfer learning, scikit-learn, TF-IDF, Logistic Regression, spaCy, LLM integration (Llama 3.1 8B, Groq API), structured JSON outputs
- **Databases & Tools**: MySQL, MongoDB, Git, Linux
- **Backend & web:** Stream lit, Pydantic v2, asyncio, Node.js, Express.js, React.js, REST APIs, HTML, CSS
- **Core Concepts**: Data Structures & Algorithms, OOP, DBMS
- **Soft Skills**: CognitiveAgility, Solution-Oriented Mindset, Execution Focus, Learning Adaptability, Ownership Mentality

PROJECTS

RAG Guard AI | Word-Level Uncertainty Scoring for Hallucination Detection in RAG Oct 2026 – Present

- This code implements a Retrieval-Augmented Generation (RAG) framework, combining a semantic search of documents with evidence-based verification to detect and flag 'hallucination' and 'unsupported' AI-generated content.
- We apply uncertainty scoring using retrieval relevance, Natural Language Inference (NLI), and evidence from semantic space to score generated content as supported, uncertain or contradicted.
- A confidence-aware evaluation is implemented to achieve 0.82 AUROC in preliminary sentence-level experiments on the RAGTruth benchmark.
- **Tech:** Python, NLP, RAG, LLMs, Sentence Transformers, DeBERTa, FAISS, NLI, Streamlit, PyTorch, RAGTruth.

Smart Park AI | Intelligent Parking Occupancy Detection System | Link Sep 2026 – Nov 2026

- Combined vehicle detection with a YOLO algorithm with parking space classification using a transfer-learned ResNet18 classifier to distinguish between BUSY and FREE parking spaces.
- Implemented Hungarian algorithm-based vehicle-to-slot matching, spatial analysis based on IoU (of bounding boxes) and a Bayesian fusion strategy for robust occupancy detection.
- Built the Streamlit dashboard for the image and/or video analysis and parking space analytics that achieved 89.75% cross-camera accuracy and 99.86% intra-scene slot-disjoint accuracy.
- **Tech:** Python, PyTorch, Torchvision, YOLO, ResNet18, OpenCV, Streamlit, NumPy, Pandas, Computer Vision

Intelligent E-Governance Grievance Center | LLM Application | Link May 2026 – Jul 2026

- Built an LLM pipeline that turns free-text citizen grievances into structured analyses: department routing, severity, priority score, sentiment, escalation risk and SLA assignment.
- Implemented Pydantic v2 schema validation and workflow verification checks to ensure reliable, structured grievance data.
- Built an asynchronous processing pipeline and Streamlit dashboard for complaint review and automated response letters, while profiling latency and throughput and achieving 90.7% validation accuracy
- **Tech:** Python, Groq API, Llama 3.1 8B, Streamlit, Pydantic v2, asyncio, Git, GitHub

TRAINING

Summer Training June 2025 – July 2025

Data Structures and Algorithms **| Link**

- Achievements: Developed an AI-powered Crop Disease Prediction System, which includes interactive visualization, confident predictions and weather-assisted disease analysis using HTML, CSS and JavaScript for the front-end.
- Key Learnings: Core Data Structures & Algorithms have been put to good use to optimize and facilitate data processing, searching and in general the application's overall performance, and to improve problem solving skills.
- Technical Exposure: In-depth working in C++, HTML, CSS, JavaScript, Git, and Algorithm designing in the assignments and the implementation of the project.

CERTIFICATES

- Oracle Agentic AI Certified Foundations Associate | Oracle — Sep 2026
- The Bits and Bytes of Computer Networking | Google (Coursera) — Sep 2024

ACHIEVEMENTS

- Solved 104 algorithmic problems on LeetCode across three difficulty levels, strengthening proficiency in data structures, algorithms, and computational problem-solving| Profile

EDUCATION

- **Lovely Professional University |** August 2023 – 2027
  Bachelor of Technology in Computer Science and Engineering; CGPA: 6.61 — Punjab, India
- **Narayana Junior college |** August 2022 – 2023
  Higher Secondary; Percentage: 83.3% — Andhra Pradesh, India
- **Lotus High School |** August 2020 –2021
  Matriculation; Percentage: 99% — Andhra Pradesh, India